Fiscal Year:	FY 2018	**Task Last Updated:**	FY 08/30/2018
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	Investigation of Host-Pathogen Interactions, Conserved Cellular Responses, and Countermeasure Efficacy During Spaceflight using the Human Surrogate Model Caenorhabditis elegans		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Immunology		
Space Biology Special Category:	None		
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City:	Tempe	**State:**	AZ
Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	03/23/2017	**End Date:**	03/31/2018
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:	ISS		
Flight Assignment:	ISS--Flown on SpaceX-5 ; ISS Expedition 41 NOTE: Extended to 3/31/2018 from extension to 12/31/2017, per F. Hernandez/ARC (Ed., 1/8/2018) NOTE: End date changed to 12/31/2017 from original end date of 10/01/2017 per F. Hernandez/ARC (Ed., 9/21/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Alverdy, John M.D. (University of Chicago) Ott, C. Mark Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	NNX17AJ53G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Investigation of Host-Pathogen Interactions, Conserved Cellular Responses, and Countermeasure Efficacy During Spaceflight using the Human Surrogate Model Caenorhabditis elegans," grant NNX10AO52G. Changes that occur to both the host immune system and pathogenesis of microbes during spaceflight represent a formidable challenge to the successful transition from short-to-long-duration spaceflight. This is a critical concern, since a) the immune system of the crew is dysfunctional during spaceflight, b) spaceflight uniquely increases the virulence and globally alters gene expression of the human pathogen Salmonella typhimurium when the bacteria are recovered post-flight and subsequently used to infect ground-based animal models, c) spaceflight-induced increases in S. typhimurium virulence are regulated by media ion composition, d) phosphate ion is sufficient to alter related pathogenesis responses in a ground-based spaceflight analogue model, and e) the evolutionarily conserved bacterial RNA chaperone protein, Hfq, serves as a master molecular regulator of many of these responses. While spaceflight has been shown to induce changes that can independently affect the host or the pathogen in a manner that is directly relevant to the development of infectious disease during spaceflight, none of these studies have been done when both the host and pathogen are simultaneously exposed to the spaceflight environment. To address this knowledge gap, we propose to challenge the human surrogate model Caenorhabditis elegans with S. typhimurium during spaceflight to study the role of the intestine in host-pathogen interactions in real-time, define virulence mechanisms, identify evolutionarily conserved responses, and test novel therapeutic strategies to prevent infectious disease. We hypothesize that the combination of a spaceflight-induced increase in virulence of the pathogen coupled with a blunted immune response of the host will result in a synergistic effect on the host-pathogen interaction such that the risk of infectious disease during spaceflight is exacerbated. We further propose that these responses represent evolutionarily conserved mechanisms at the level of both host and pathogen that are dependent on media ion composition, specifically the local concentration of phosphate. See also: http://www.nasa.gov/
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will utilize the microgravity spaceflight platform 1) to broaden our knowledge of the host-pathogen interaction that leads to infectious disease, and 2) for the development of new therapeutic strategies to combat infectious disease for the general public.
Task Progress:	SUPPLEMENTAL REPORTING FOR FINAL REPORT, AUGUST 2018 Completed RNA-Seq gene expression profiles for C. elegans spaceflight and ground control cultures infected with Salmonella cultured in either Lennox Broth (LB) or LBM9 media (as well as flight and ground non-pathogenic E. coli controls cultured in the same media). RNA-Seq analysis was successfully performed (for both host and microbe) using high integrity RNA extracted from samples. Our findings indicate significant differences in gene expression between flight and ground samples. Manuscript being finalized for publication. ANNUAL REPORTING, JANUARY 2018 • Completed virulence analysis of C. elegans spaceflight and identical ground control cultures infected with Salmonella cultured in LB-PO4 (as well as spaceflight and ground non-pathogenic E. coli controls cultured in the same media). Currently finalizing analysis of Pi-PEG virulence video data. • RNA-Seq gene expression profiles are currently being analyzed for C. elegans spaceflight and ground control cultures infected with Salmonella cultured in LB and LBM9 media (as well as flight and ground non-pathogenic E. coli controls cultured in the same media). Manuscript is in preparation.
Bibliography Type:	Description: (Last Updated: 07/02/2025)